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Manufacturing manuals: index

Start with **STANDARD_PROCEDURES.md** (PCB ordering, SMT/THT assembly, soldering, programming, 3D printing, laser cutting, QA, packaging, compliance, troubleshooting) and the templates. Each product has a `MANUFACTURING_MANUAL.md` with its bill of materials, build steps, calibration, end-of-line test and limits.

Honest status: nothing in this repository has been built, assembled or tested as hardware. 'Order-ready' means the automatic checks pass (KiCad DRC 0 errors, schematic equals layout, every part has a verified LCSC number, Gerber/BOM/CPL consistent): it does **not** mean an engineer has reviewed the layout. Treat the first batch as a prototype run.

Utility tools

Project	Manual	Ordering	Board	Order confidence	Works as intended	Main caveat
Seeti Counter	manual	ordering	**order-ready**	Medium-High	Medium	firmware compiled, not run on hardware
Gas Taraju (LPG cylinder scale)	manual	ordering	**order-ready**	Medium-High	Medium	load-cell calibration per unit
Fridge Door Guard	manual	ordering	**order-ready**	Medium-High	Medium-High	
Bijli Meter Reader	manual	ordering	**order-ready**	High	Medium	
Bijli Log	manual	ordering	**order-ready**	High	Medium	OLED/RTC drivers untested
Chulha Timers	manual	ordering	**order-ready**	Medium-High	Medium-High	display driver untested
Rain Alert Bell + plate	manual	ordering	**order-ready**	Medium-High	Medium	plate: `rainplate` board
Tap Litre Meter	manual	ordering	**order-ready**	Medium-High	Medium	needs bucket calibration
Masala Rack (laser-cut)	manual	ordering	no PCB	Medium-High	Medium-High	kerf coupon first
Laptop Stand (laser-cut)	manual	ordering	no PCB	Medium-High	Medium-High	stability limit stated

Products

Project	Manual	Ordering	Board	Order confidence	Works as intended	Main caveat
1 Desi Games Studio (print)	manual	ordering	no PCB	Medium-High	Medium	language + fact review before print
2 Kalakriti Automata (laser)	manual	ordering	no PCB	Medium-High	Medium	frame layouts provisional
3 Prayog STEM lab kit	manual	ordering	**order-ready**	High	Medium-High	sensors uncalibrated

Project	Manual	Ordering	Board	Order confidence	Works as intended	Main caveat
4 Jal-Rakshak tank controller	manual	ordering	**order-ready**	Medium-High	Medium	**mains version cannot be made**: low-voltage prototype only
5 Kheti-Mitra irrigation	manual	ordering	**order-ready**	Medium-High	Medium	field trial required
6 Gas-Suraksha leak alarm	manual	ordering	**order-ready**	High	Low-Medium	**not certified, not for sale**

Toys (10)

Project	Manual	Ordering	Board	Order confidence	Works as intended	Main caveat
01 Reflex Raja	manual	ordering	**order-ready**	Medium-High	Medium-High	shares the Desi Arcade board
02 Taal Memory	manual	ordering	**order-ready**	Medium-High	Medium-High	shares the Desi Arcade board
03 Aloo Bomb	manual	ordering	**order-ready**	Medium-High	Medium-High	shares the Desi Arcade board
04 Beat Runner	manual	ordering	**order-ready**	Medium-High	Medium-High	
05 Rangoli Spiro Gears	manual	ordering	no PCB	Medium-High	Medium-High	kerf test first
06 Paper Rocket Launcher	manual	ordering	no PCB	Medium-High	Medium	projectile toy: lab review before sale; no drawings
07 Tilt Maze	manual	ordering	no PCB	Medium-High	Medium	gimbal drawn and checked; cut one test set
08 Rubber-band Racer	manual	ordering	no PCB	Medium-High	Medium-High	provisional cut file
09 Marble Cricket	manual	ordering	no PCB	Medium-High	Medium	provisional cut file; calibration drop
10 Katha Cubes	manual	ordering	no PCB	Medium-High	Medium-High	illustrations and proofreading missing

Bio / agri / civil / robotics programmes

Project	Manual	Ordering	Board	Order confidence	Works as intended	Main caveat
KINETIC motion controller + vibration puck	manual	ordering	**order-ready**	High	Medium	never run on a motor; AI on synthetic data
MITTI-NETRA soil + structure node	manual	ordering	**order-ready**	High	Medium	modem is a plug-in breakout
BioTray vertical-farm controller	manual	ordering	**order-ready**	Medium-High	Medium	calibration values are placeholders
BioWatt MFC harvester	manual	ordering	**order-ready**	Medium-High	Low-Medium	cold start needs 600 mV; marginal on one MFC
GEONODE foundation monitoring	manual	ordering	no PCB	Medium	Low-Medium	synthetic data; reuses MITTI-NETRA board
AQUAPONICS controller	manual	ordering	no PCB	Medium	Low-Medium	runs on BioTray; biological limits are placeholders

Project	Manual	Ordering	Board	Order confidence	Works as intended	Main caveat
SORTER bio-robotic platform	manual	ordering	no PCB	Medium	Low	vision baseline only; uses KINETIC
AGROVER solar rover	manual	ordering	no PCB	Medium	Low-Medium	dead-reckoning only; safety review needed

Other projects

Project	Manual	Ordering	Board	Order confidence	Works as intended	Main caveat
Jal-Netra (engineering marvel)	manual	ordering	**order-ready**	Medium-High	Low-Medium	**Gate 1 acoustic test first; not sellable**
ASTRADAND gesture wand (sci-fi project)	manual	ordering	**order-ready**	Medium	Medium	software-verified; never built; Hindi names unreviewed
Gravity toys (5)	manual	ordering	no PCB	Medium-High	Medium	all 5 drawn; Drop Quest has 60 verified levels
TRISHNA urinal hydration monitor	manual	ordering	**order-ready**	Medium-High	Low-Medium	reagentless SG estimate needs per-user refractometer calibration; wellness trend only, not a medical device
Wave 1 electronics toys	manual	ordering	**order-ready**	Medium-High	Low-Medium	AP1 board order-ready; MP3 module and 4-digit display module are bought-in
Prototype packs (10)	manual	ordering	no PCB	Medium	Medium	bench prototypes
DJ / party system	manual	ordering	no PCB	Medium	Low-Medium	programme plan; phase-0 prototypes D1/D2 in prototypes/

Confidence columns: see `costing/CONFIDENCE.md` for the rubric and the reason behind every rating. Ordering how-to for any project: `ORDERING_GUIDE.md`. Priced bills of materials: `costing/BOM_COSTS.md`.